Journal of Magnetism and Magnetic Materials 473 (2019) 42–50

Contents lists available at ScienceDirect

Journal of Magnetism and Magnetic Materials

journal homepage: www.elsevier.com/locate/jmmm

Research articles

Numerical study of biomagnetic fluid flow in a duct with a constriction affected by a magnetic field

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ARTICLE INFO

Keywords:
Biomagnetic fluid
Magnetic field
Fluid dynamics
Wall shear stress

ABSTRACT

In this paper, the influence of a non-uniform magnetic field on biomagnetic fluid flow in a duct with a constriction is investigated numerically. The magnetic field is generated by a wire carrying electric current located outside the duct. The biomagnetic fluid dynamics model, which is based on the ferrohydrodynamics and magnetohydrodynamics, is used to study the fluid flow, and a computational grid properly covering the magnetic field is used for the simulation. In this investigation, the influence of magnetic field on biomagnetic fluid flow in various percentages of the duct constriction, and the effect of the magnetic field on the biomagnetic fluid flow in various strengths of magnetic force are studied. The magnetic field is spatially varying, and the magnetic field strength decreases by increasing the distance from the wire carrying electric current which is the magnetic source. The results show that, the influence of the magnetic field on biomagnetic fluid flow is considerable near the magnetic source. Applying the magnetic field with enough strength causes the recirculation areas downstream of constriction to become smaller. In addition, applying the magnetic field has significant effects on the wall shear stress, so that it can decrease the shear stress on the wall away from the magnetic source.

1. Introduction

Biomagnetic fluids are biological fluids which are affected by magnetic field. Understanding about the behavior of biomagnetic fluids, the most distinguished magnetic fluid is blood, under the influence of magnetic field has been considered by numerous researchers. Also, the behavior of the magnetic field on the various geometries in industrial tools has been respected by the researchers [1–6]. Up to now, numerous applications about this field have been presented in the engineering and medical sciences including cell separation, targeted drug delivery; wound bounding due to the magnetic field effects and curing cancer via magnetic hyperthermia [7–10]. The red cells of blood are consisted of protein called hemoglobin which is responsible for carrying oxygen inside the human body. The investigations reveal that, the red cells of blood orientate in the presence of magnetic field [11,12]. The constriction in some parts of vessels causes the creation of recirculation flows and high values of shear stress in these areas. Also, lower and oscillating values of walls shear stress around the downstream of constricted duct develops the creation of intimal thickening [13].

According to (Fig. 1) [14], Atherosclerosis is an arterial illness causing incremental constriction of vessel (lumen) and hardening of vessels walls due to the sedimentation of cholesterol on the internal walls of vessels. By developing the sedimentations on vessels walls, a plaque is created causing a protuberance in lumen and limiting the blood flow. Eventually, plaques may completely block the lumen or a thrombus may block the constricted vessel. For these reasons, researchers have investigated the biomagnetic fluid flow in different sections with specific geometrical conditions. Khashan and Haik [15] studied the effect of magnetic field of a magnet on biomagnetic fluid flow downstream of eccentric stenotic orifice, two-dimensionally.

In their study, the magnetic source was placed near the wall away from the inlet orifice of flow and indicated that, using magnetic field causes growing of vortex and the postponement in the reattachment point. Also, by increasing the magnetic force, the returned zone and the reattachment points tend toward the downstream. Tzirtzilakis [16] investigated the biomagnetic fluid flow inside a symmetrical stenosis

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https://doi.org/10.1016/j.jmmm.2018.10.043
Received 9 May 2018; Received in revised form 25 September 2018; Accepted 9 October 2018
Available online 10 October 2018
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Nomenclature

(a, b)	place of the wire in x-y plan
B_0	magnetic field induction at the point (*a*, h) (T)
$\vec{B}$	magnetic field induction (T)
B_x	magnetic field induction component in x direction (T)
B_y	magnetic field induction component in y direction (T)
h	channel height (m)
H	magnetic field intensity (A/m)
H_0	magnetic field intensity at the point (*a*, h) (A/m)
H_x	magnetic field intensity component in x direction (A/m)
H_y	magnetic field intensity component in y direction (A/m)
Ha	Hartmann number ($B_0 h\sqrt{\sigma/\mu}$)
I	electric current intensity (A)
$\vec{J}$	electric current density (A/ m^2)
M	magnetization (A/m)
Mn	Magnetic number ($\chi B_0^2/\mu_0 \rho \bar{U}_0^2$)
N	Stuart number ($B_0^2 h\sigma/\rho\bar{U}_0$ = Ha^2/Re)
P	pressure (Pa)
Re	Reynolds number ($\rho\bar{U}_0 h/\mu$)
$\vec{V} = (u, v)$	velocity field (m/s)
$\bar{U}_0$	average inlet velocity (m/s)
X	axis in Cartesian coordinates
Y	axis in Cartesian coordinates

Greek Letters

δ	Channel height tightness (m)
μ	dynamic viscosity (kg/m s)
μ0	magnetic permeability of vacuum ($=4\pi \times 102^{-7}$ Tm/A)
ρ	density (kg/m^3)
σ	electrical conductivity (s/m)
τ	shear stress (N/m^2)
Γ	magnetic susceptibility

Superscripts

*	dimensionless symbol

duct affected by magnetic field caused by crossing electric current from wire, two-dimensionally. He indicated the creation of two symmetric vortexes downstream of constricted duct with pure hydrodynamic flow. By applying the magnetic field, the vortex close to the magnetic source grows up and consequently, the reattachment point of flow transfers toward the downstream. His results revealed that, in some cases, by applying magnetic field and the growth of vortex near the magnetic source, the other vortex created in hydrodynamic state, becomes smaller and even by increasing the strength of magnetic field, this vortex vanishes and the flow separation does not occur on one of the two plates.

Mustapha et al. [17] studied the effect of uniform magneto-hydrodynamic field on blood flow inside an irregular multi-stenosed artery, two-dimensionally and axially symmetric. They revealed that, with the absence of magnetic field, the maximum value of shear stress of wall is related to the stenosis sections and by applying the magnetic field, the shear stress of wall rises and by increasing the magnetic force, stress rises, too. They also concluded that, by applying magnetic field, the flow separation can be controlled significantly. So that, the size of separated areas decreases as the magnetic force increases and in higher strength of magnetic field, these areas are vanished completely.

Tzirtzilakis et al. [18] investigated the biomagnetic fluid flow in a three-dimensional duct with magnetic field caused by crossing electric current from wire parallel with the longitudinal axis of duct. He denoted that, the augmentation of magnetic field strength causes the creation of two vortexes related to the secondary flow and lagging of axial flow. By increasing the magnetic field strength, especially near to the magnetic source, this phenomenon is more significant. They figured out that, the magnetic field has considerable influence on shear stresses of duct walls, especially wall stresses near the magnetic source.

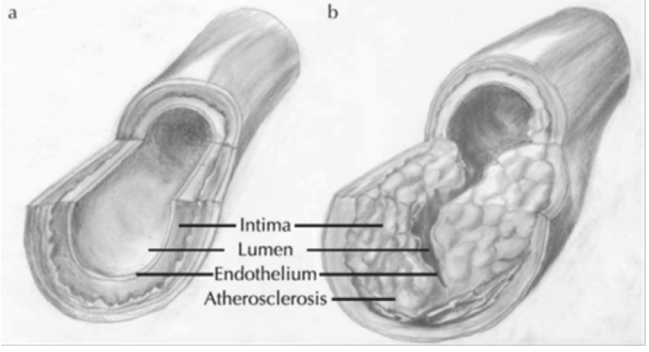


Fig. 1. Normal vessel wall (left) and Atherosclerotic thickening of the vessel wall (right) [14].

Papadopoulos and Tzirtzilakis [19] studied the biomagnetic fluid flow inside a curved square duct with magnetic field caused by crossing electric current from wire parallel with the longitudinal axis of duct. They showed that, the position of magnetic field plays an important role in the formation of flow field. Based on the position of magnetic field, the created vortexes on the transverse plate can change in their number and asymmetric figure. Kenjeres et al. [20] investigated the blood flow in realistic arteries with the presence of magnetic field caused by the crossing electric current from wire and indicated that, the considerable changes of local pressure happen in areas with considerable strength of magnetic field. Tzirtzilakis and Xenos [21] studied the biomagnetic fluid flow in a driven cavity under the influence of magnetic field caused by the crossing electric current from wire and revealed that, when Lorentz force is not considered, applying magnetic field causes the creation of additional vortex, comparing to the state with pure hydrodynamics and the effect of magnetic field is local and near the magnetic source. Mousavi et al. [22] indicated that, with the presence of magnetic field caused by wire carrying electric current, in different computational grids, results have oscillating behavior and the different figures of the magnetic field in different computational grids cause this oscillating behavior. They found that, for achieving the accurate results, the computational grids which properly cover the magnetic field must be used. Sankar and Lee [23] studied the effect of uniform magnetic field on the pulsatile blood flow inside the mild stenosis arteries with non-Newtonian blood fluid. They indicated that, in the presence of magnetic field, the skin friction increases significantly and the augmentation of magnetic force causes the increment of skin friction. The use of nanofluids in various geometries and small scale ducts and biological nanofluids in arterial ducts has been investigated by researchers in recent studies [24–32]. Thus, researchers are investigating the behavior of newtonian and Non-newtonian nanofluids [33–37] in geometries such as microchannels, and try to simulate the behavior of nanofluidic flows in these geometries [38–42]. In the present paper, the effect of the magnetic field caused by wire carrying electric current, which is a non-uniform magnetic field, on the biomagnetic fluid flow in a two-dimensional duct with constriction has numerically been investigated using finite volume method. In this research, the influence of the magnetic field on the recirculation areas and the distribution of shear stress have been studied in different percentages of constriction. In current study, the biomagnetic fluid dynamic model based on the ferrohydrodynamics and magnetohydrodynamics has been used, and for the simulation of the effects of this

magnetic field, in that computational girds occupy center stage in the accuracy of results [22], a computational grid that properly covers the magnetic field has been employed.

2. Problem statement and dimensioning

In the present numerical simulation, the flow is two-dimensional, steady, laminar and incompressible and the biomagnetic fluid is Newtonian, homogeneous and electric conductor. The fluid flow is considered inside a duct with constriction under the influence of the magnetic field generated by a wire carrying electric current. (Fig. 2) indicates the schematic of the studied problem. The inlet height of the duct is shown by h and the wire carrying electric current is normal to x-y plane shown as a point with the coordinates of (a, b) (a = 7 h and b = 1.2 h).

The curvature of the duct is based on Eq. (1).

$$y = \frac{\delta}{2}\left(1 + \cos\left(\frac{(x-7h)\pi}{h}\right)\right) \qquad 6h \leqslant x \leqslant 8h \tag{1}$$

where, δ is the maximum height of the constriction. The viscosity changes due to the applying of magnetic field are neglected. The equilibrium magnetization is considered (where $\vec{M}$ and $\vec{H}$ are parallel with each other) and the continuation of magnetic induction $\nabla . \vec{B} = 0$ is considered.

3. Governing equations and solving procedure

According to the mentioned assumptions, the governing equations on the flow (Eqs. (2)–(4)) are continuity and momentum equations along the x and y axes as follows [43]:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \tag{2}$$

$$\rho\left(u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y}\right) = -\frac{\partial p}{\partial x} + \mu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) + \mu_0 M\frac{\partial H}{\partial x} + \sigma(vB_xB_y - uB_y^2) \tag{3}$$

$$\rho\left(u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y}\right) = -\frac{\partial p}{\partial y} + \mu\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right) + \mu_0 M\frac{\partial H}{\partial y} + \sigma(uB_xB_y - vB_x^2) \tag{4}$$

On the bottom and top walls of the duct, the no-slip boundary condition is applied and in the inlet of the duct (x = 0), the fluid flow enters as completely developed and laminar with the distribution presented in Eq. (5):

$$u(y) = \frac{3}{2}\bar{U}_0\left(1 - 4\left(\frac{y - h/2}{h}\right)^2\right) \tag{5}$$

where, $\bar{U}_0$ is the average inlet velocity. In the outlet of the duct (x = 22 h), the diffusion fluxes normal to the outlet plane are considered zero for all of the flow parameters. In these equations, $\vec{V} = (u, v)$ is the velocity field, ρ is the density of the biomagnetic fluid, P is the pressure, μ is the dynamic viscosity of the biomagnetic fluid and μ_0 is the magnetic permeability of vacuum. In addition, M indicates the magnetization, H is the magnetic field intensity, $\vec{B}$ is the Magnetic field induction and σ is the electrical conductivity of the biomagnetic fluid. In Eqs. (3) and (4), $\mu_0 M\frac{\partial H}{\partial x} + \sigma(vB_xB_y - uB_y^2)$ and $\mu_0 M\frac{\partial H}{\partial y} + \sigma(uB_xB_y - vB_x^2)$ are the components of the magnetic force per unit volume along the x and y, respectively. $\mu_0 M\frac{\partial H}{\partial x}$ and $\mu_0 M\frac{\partial H}{\partial y}$ are due to magnetization that depends on the existence of the magnetic gradient. The terms of $\sigma(vB_xB_y - uB_y^2)$ and $\sigma(uB_xB_y - vB_x^2)$ indicate the components of Lorentz force per unit volume (Lorentz force per unit volume equals $\vec{J} \times \vec{B}$, where $\vec{J}$ is the density of electric current, and $\nabla \times \vec{H} = \vec{J} = \sigma(\vec{V} \times \vec{B})$) caused by electrical conductivity of fluid. The relation between the magnetic field induction and the magnetic field intensity is as Eq. (6).

$$\vec{B} = \mu_0 \vec{H} \tag{6}$$

The magnetization is considered as the linear Eq. (7) [43],

$$M = \chi H \tag{7}$$

where, χ is the magnetic susceptibility of the biomagnetic fluid and it is constant. The components of the magnetic field intensity along the x and y axes, H_x and H_y [22], are as Eq. (8).

$$H_x = \frac{I}{2\pi}\frac{y-b}{(x-a)^2 + (y-b)^2}$$
$$H_y = \frac{I}{2\pi}\frac{x-a}{(x-a)^2 + (y-b)^2} \tag{8}$$

where, (a, b) is the location of the wire and I indicates the electric current intensity. Eq. (9) presents the magnitude of the magnetic field intensity.

$$H = \frac{I}{2\pi}\frac{1}{\sqrt{(x-a)^2 + (y-b)^2}} \tag{9}$$

The dimensionless variables are defined as Eq. (10).

$$x^* = \frac{x}{h},\ y^* = \frac{y}{h},\ u^* = \frac{u}{\bar{U}_0},\ v^* = \frac{v}{\bar{U}_0},\ p^* = \frac{p}{\rho\bar{U}_0^2},\ H_x^* = \frac{H_x}{H_0},\ H_y^* = \frac{H_y}{H_0} \tag{10}$$

By substituting the dimensionless variables in Eqs. (2)–(4), Eqs. (11)–(13) are achieved which are the dimensionless forms of Eqs. (2)–(4).

$$\frac{\partial u^*}{\partial x^*} + \frac{\partial v^*}{\partial y^*} = 0 \tag{11}$$

$$u^*\frac{\partial u^*}{\partial x^*} + v^*\frac{\partial u^*}{\partial y^*} = -\frac{\partial p^*}{\partial x^*} + \frac{1}{\mathrm{Re}}\left(\frac{\partial^2 u^*}{\partial x^{*2}} + \frac{\partial^2 u^*}{\partial y^{*2}}\right) + \mathrm{Mn}H^*\frac{\partial H^*}{\partial x^*} + \mathrm{N}(v^*H_x^*H_y^* - u^*H_y^{*2}) \tag{12}$$

$$u^*\frac{\partial v^*}{\partial x^*} + v^*\frac{\partial v^*}{\partial y^*} = -\frac{\partial p^*}{\partial y^*} + \frac{1}{\mathrm{Re}}\left(\frac{\partial^2 v^*}{\partial x^{*2}} + \frac{\partial^2 v^*}{\partial y^{*2}}\right) + \mathrm{Mn}H^*\frac{\partial H^*}{\partial y^*} + \mathrm{N}(u^*H_x^*H_y^* - v^*H_x^{*2}) \tag{13}$$

The dimensionless numbers [21] are defined as follows:

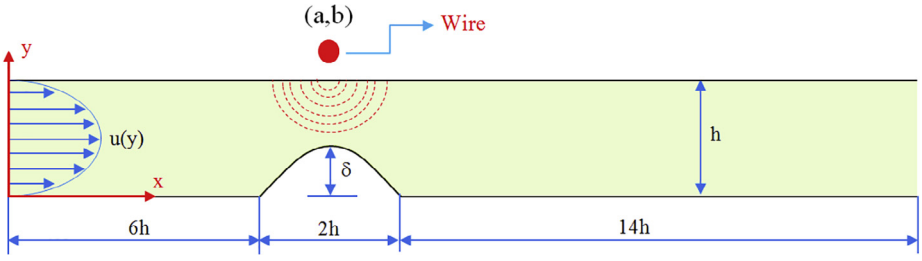


Fig. 2. The schematic diagram of physical model.

$$\mathrm{Re} = \frac{\rho \bar{U}_0 h}{\mu}, \quad \mathrm{Mn} = \frac{\mu_0 \chi H_0^2}{\rho \bar{U}_0^2} = \frac{\chi B_0^2}{\mu_0 \rho \bar{U}_0^2} \tag{14}$$

$$\mathrm{N} = \frac{\mu_0^2 H_0^2 h \sigma}{\rho \bar{U}_0} = \frac{B_0^2 h \sigma}{\rho \bar{U}_0} = \frac{\mathrm{Ha}^2}{\mathrm{Re}} \tag{15}$$

where Re is Reynolds number, Mn is the magnetic number and N is Stuart number which equals with the ratio of square Hartmann number (Ha) to Reynolds number. H_0 and B_0 are respectively the magnetic field intensity and magnetic field induction at (a, b) and $B_0 = \mu_0 H_0$. The changes of shear stress on walls can be calculated by following equation:

$$\tau_{yx} = \mu \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) \Big|_{On\ \ the\ wall} = \mu \frac{\partial u}{\partial y} \Big|_{On\ \ the\ wall} = \frac{\mu \bar{U}_0}{h} \frac{\partial u^*}{\partial y^*} \Big|_{On\ \ the\ wall} \tag{16}$$

For solving the governing equations on this problem, the finite volume method has been used. For coupling the velocity and pressure, SIMPLE algorithm has been employed. The convective and diffusive terms in momentum equations have been discretized by using the second order upwind method.

4. Independency from the computational grids

In current numerical study, in order to have an appropriate computational grid for simulating the effect of the magnetic field caused by wire carrying electric current on the magnetic fluid flow, the non-uniform computational grids have been used (Fig. 3). According to (Fig. 3), in order to investigate the independency from the computational grid and appropriate covering the magnetic field and avoiding the use of grids with high cell numbers causing the increase of computational volume, the non-uniform grids 1920 × 225, 2400 × 275, 2880 × 250, 2400 × 225 and 2400 × 275 have been considered. The independency from the computational grids has been studied for 50% constriction (δ/h = 0.5). In (Fig. 4), the results related to the changes of dimensionless velocity along x versus the changes of y^* at $x^* = 8$ are shown. By changing the number of grids along x according to Figure (a-4), the number of optimized grids along this axis is 2400. After achieving the optimized grid number along x, this grid number is assumed constant and according to Figure (b-4), the changes of grid number have been carried out along y. Based on the obtained results, the non-uniform grid number of 2400 × 250 has been chosen as the appropriate grid number.

5. Results and discussions

In order to validate the present study, the results obtained from the numerical study have been compared with the results of Khashan and Haik [15]. Khashan and Haik [15] investigated the influence of magnetic field on biomagnetic fluid flow inside an eccentric stenosis orifice, two-dimensionally. In their study, by placing a steady magnetic source close to the wall which was away from the inlet orifice of flow, they investigated the hydrodynamic behavior of biomagnetic flow using finite volume method. The results of this comparison have been validated for determining the friction coefficient on the bottom wall for various local conditions in the presence of magnetic field (Mn = 500) and absence of magnetic field (Mn = 0) in Reynolds number of 50. According to (Fig. 5), the assumptions of present numerical solving have proper accuracy and negligible error.

In contours of (Fig. 6), the streamlines are indicated for 50% constriction (δ/H = 0.5) in various strengths of magnetic field. As it is observed, in the absence of the magnetic field, a recirculation area is created downstream of the constricted area, which is because of the positive pressure gradient and flow separation in this area. By applying the magnetic field, this recirculation area becomes small due to the downward flow, and by increasing the magnetic force, this recirculation area becomes smaller, indicating that using magnetic field with proper power in proper location controls the vortexes. As it is seen from Figure (c-6) the magnetic force in Mn = 2.54 × 10^5 and N = 0.29502 causes the creation of two small recirculation areas close to the magnetic source. The reason of the creation of these two recirculation areas is that magnetic force accelerates the flow in the directions of $\mu_0 M \frac{\partial H}{\partial x}$ along x and $\mu_0 M \frac{\partial H}{\partial y}$ along y. These two recirculation areas become bigger in Mn = 4.0 × 10^5 and N = 0.46097 (Figure (d-6)).

In (Fig. 7), the gradient of dimensionless velocity of $\frac{\partial u^*}{\partial y^*}$ in various strengths of magnetic field on the bottom wall of duct has been shown. In fact, the changes of velocity parameters normal to the duct surface determine the shear stress. In each section where these changes are high, the shear stress rises. Due to the stenosis located in 7 h, while crossing from this section, the fluid has strong velocity gradients. Because of the continuity of fluid motion, with decreasing the surface, fluid velocity increases, for this reason the maximum gradient has been occurred in maximum constriction. By moving the fluid inside the duct, due to the influence of momentum, the fluid tends to move on the curved direction (stenosis section of duct), however, in these areas the pressure gradients are high and they hinder the fluid motion on the curved surface as an opposite force. This behavior becomes more obvious in the state with no or less magnetic force. Due to the mentioned reasons, in (Fig. 7), the changes of dimensionless velocity gradients become higher in areas before and after the constricted duct.

In the constricted section of duct, by applying the magnetic field, the shear stress decreases on the bottom wall and by increasing the magnetic force, the shear stress decreases more, which is because of the motion of flow toward the bottom wall under the influence of the magnetic force causing the reduction of the velocity gradient close to the wall. It means that, by applying stronger magnetic force, the velocity gradients of fluid decreases on the bottom surface of the duct and the fluid flow field has better coincidence with it. Based on (Fig. 8), the stress on the top wall has reversed trend compared to the stress on bottom wall. Meaning that, by applying the magnetic field, the stress rises and with increasing the magnetic force, the maximum value of

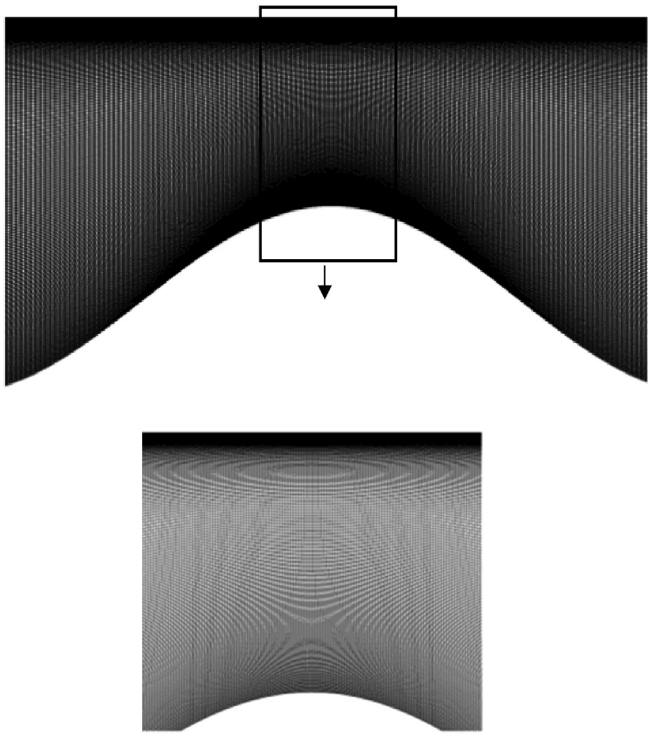
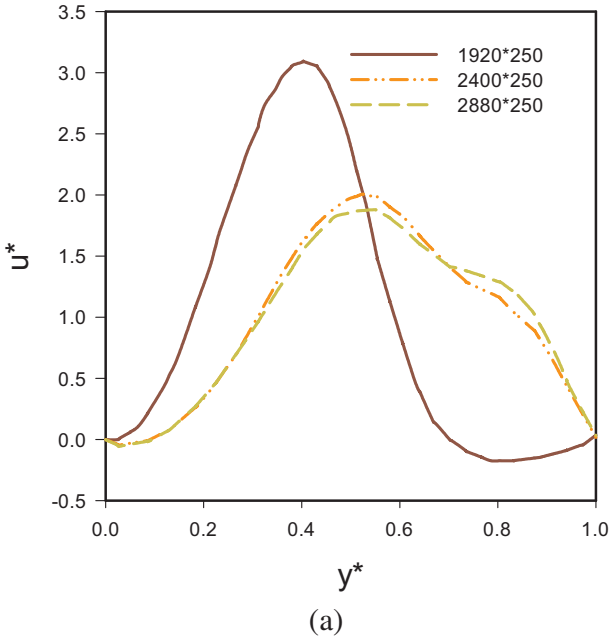


Fig. 3. Computational grid.



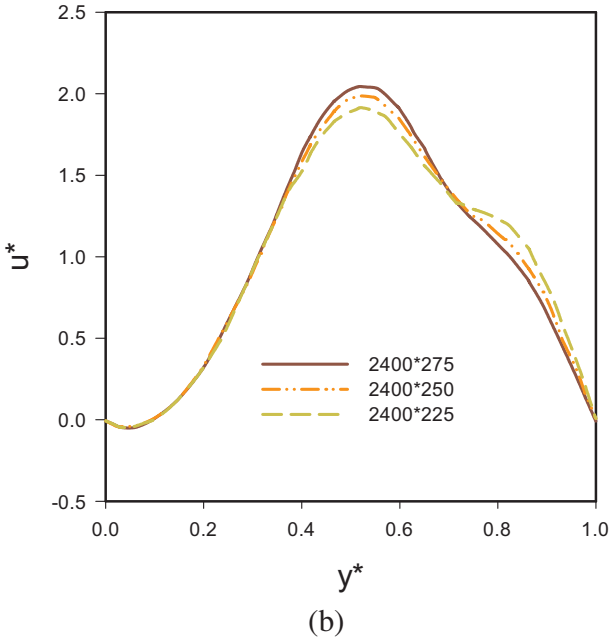


Fig. 4. Grid independency along (a) x and (b) y directions: Comparison of the dimensionless velocity component along x direction on the transverse line ($x^* = 8$) at Re = 100, Mn = 4.0 × 105 and N = 0.46097.

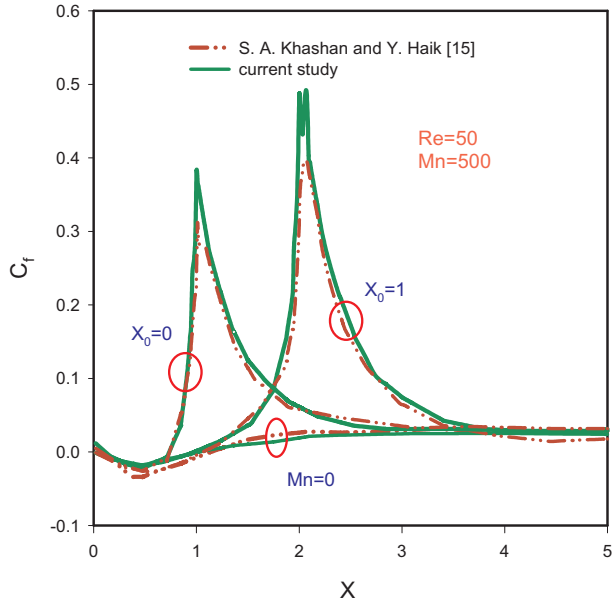


Fig. 5. Validation of present numerical study with the results of Khashan and Haik [15].

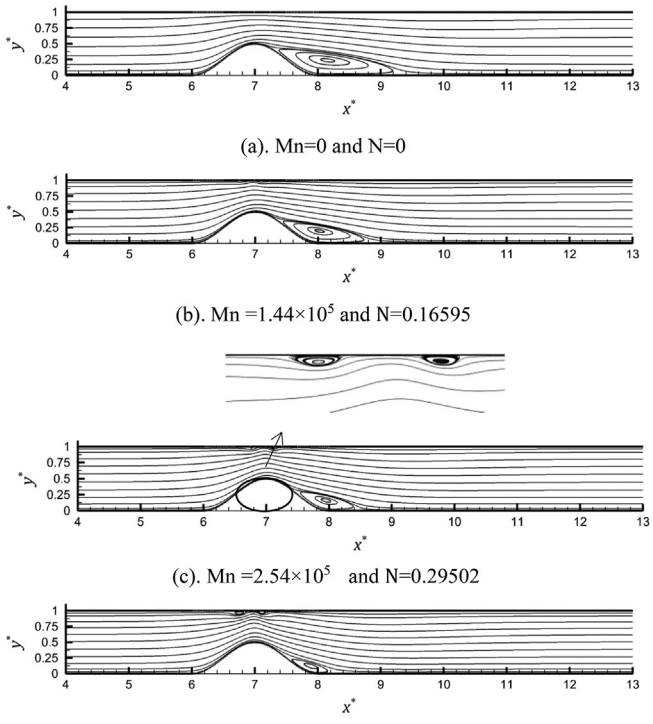


Fig. 6. Streamlines for δ/h = 0.5 in Re = 100.

stress rises too. The maximum value of stress in Mn = 4.0 × 10^5 and N = 0.46097 is almost 5.4 times the stress in state with no magnetic field. In (Fig. 8), the magnetic field causes the gradient of velocity on the top wall to change its mark several times. This is attributed to the existence of the two vortexes near the magnetic field. As it is seen in (Figs. 7 and 8), the influence of the magnetic field on the shear stress caused by the motion of biomagnetic fluid flow on duct walls, by increasing the distance from the wire carrying electric current, declines. In general, by investigating the changes of stress domain on the top and bottom walls, it can be said that the changes of stress in areas before and after the constriction has covered more areas on the bottom wall. The reason of this behavior is because of the direct influence of velocity gradients due to the collision of fluid with the curved surface of duct.

In contours of (Figs. 9 and 10), the streamlines are shown for 50% constriction (δ/H = 0.5) in the absence of magnetic field and in Mn = 4.0 × 10^5 and N = 0.46097 in Reynolds numbers of 150 and 200. In these contours, the effect of the increment of Reynolds number and the magnetic force on the behavior of streamlines is studied in $x^* = 4$ to $x^* = 13$. By increasing fluid velocity, the pressure gradients rise in the areas after the protuberance and as the forces against fluid motion cause the separation of fluid and the creation of vortex in areas close to the bottom surface of the duct. The comparison Figs. 9 and 10 reveals that the recirculation zones behind the constriction become bigger by increasing Reynolds number where the flow separation occurs

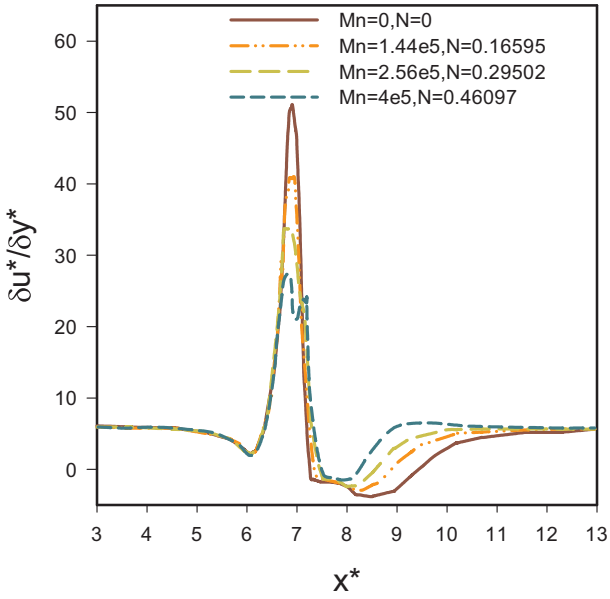


Fig. 7. Dimensionless velocity gradient on the bottom wall for δ/h = 0.5 and Re = 100.

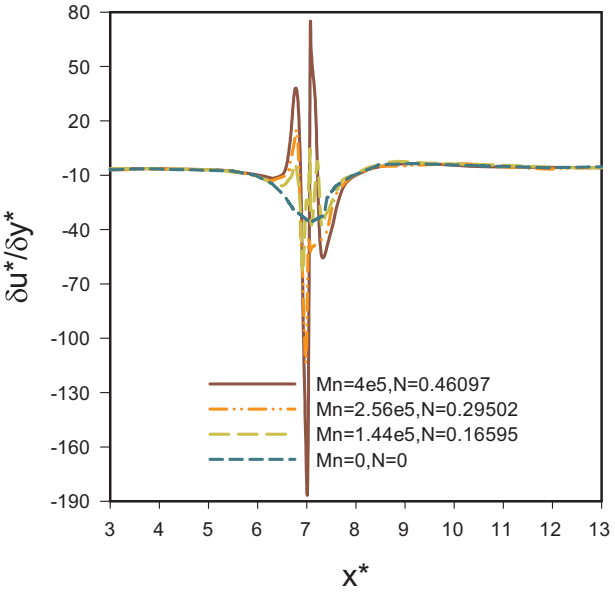
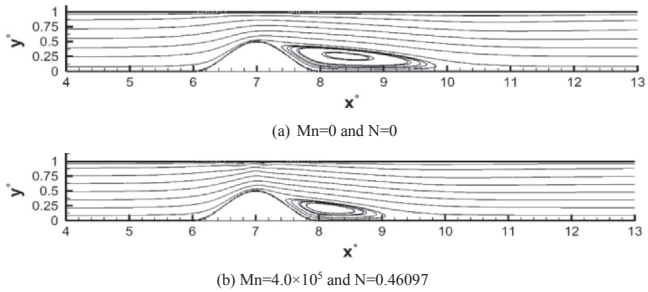


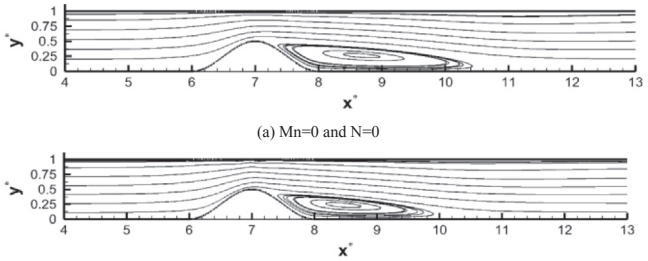
Fig. 8. Dimensionless velocity gradient on the top wall for δ/h = 0.5 and Re = 100.



(a) Mn=0 and N=0

(b) Mn=4.0×10^5 and N=0.46097

Fig. 9. Streamlines for δ/h = 0.5 and Re = 150.



(a) Mn=0 and N=0

(b) Mn=4.0×105 and N=0.46097

Fig. 10. Streamlines for δ/h = 0.5 and Re = 200.

earlier. The strong magnetic field causes the reduction of pressure gradients in areas after the constriction and with better coincidence of fluid with surface causes the reduction of wake region. Although in Reynolds numbers of 150 and 200, by applying magnetic field, the vortexes behind the constriction become smaller, the results indicate that by increasing Reynolds number, due to the improvement of flow momentum, the effect of magnetic field decreases and the areas with vortexes cover more zones.

In (Fig. 11a), the changes of dimensionless velocity gradient on the bottom wall of the duct are shown in different Reynolds numbers without the magnetic field. By increasing Reynolds number, the maximum velocity gradient rises causing the augmentation of shear stress. In fact, the changes of this figure in the constricted areas indicate that, based on the law of continuity, the value of crossing mass from the section is steady in each Reynolds number and by decreasing the section in higher Reynolds numbers, fluid velocity increases. On the other hand, the motion of fluid flow in the constricted section of duct accompanies with stronger gradients, after crossing from the constricted section, due to more depreciation of momentum, fluid has weaker velocity gradients. This behavior of the changes of fluid velocity is obvious in Reynolds numbers of 200, 150 and 100, respectively, indicating the steadiness of energy for fluid motion. In (Fig. 11b), by applying the magnetic field in each Reynolds number, the maximum velocity gradient declines. This reducing trend is more significant in lower Reynolds numbers indicating less sensibility of fluid flow to the magnetic force in higher Reynolds numbers.

In contours of (Fig. 12), the local behavior of streamlines is shown for 25% constriction (δ/H = 0.25) with and without the magnetic field. By decreasing the height of constriction, the opposite gradients of fluid motion decrease and the fluid has almost complete coincidence with surfaces and the protuberance cannot be a stimulant for flow separation. In each two states of presence and absence of the magnetic field, on the bottom wall of the duct and downstream, there is no recirculation area. On the top wall of the duct, in the presence of magnetic field, two recirculation areas are created near the magnetic source due to the flow movement in the direction of magnetic forces. It causes the change of velocity gradient in this area. This behavior shows that the direction of magnetic field tends to stick the fluid to the bottom surface and the effect of this force on the top surface causes flow separation and the creation of vortexes.

According to (Fig. 13a), the dimensionless velocity gradient in this graph in comparison to the state of δ/H = 0.5, in the same Reynolds number, decreases significantly. The reason of this behavior is that the reduction of constriction causes less velocity gradients. Based on the figures, on the bottom wall, the maximum gradient of dimensionless velocity happens in the maximum constriction, and by applying the magnetic field this gradient declines. In (Fig. 13b), the dimensionless velocity gradients on the top wall are shown. In the presence of magnetic field, two vortexes are created close to the magnetic source, and consequently the shear stress changes dramatically on the top wall. By applying magnetic field, the maximum velocity gradient and shear stress are increased. Moreover, due to the existence of two vortexes, the

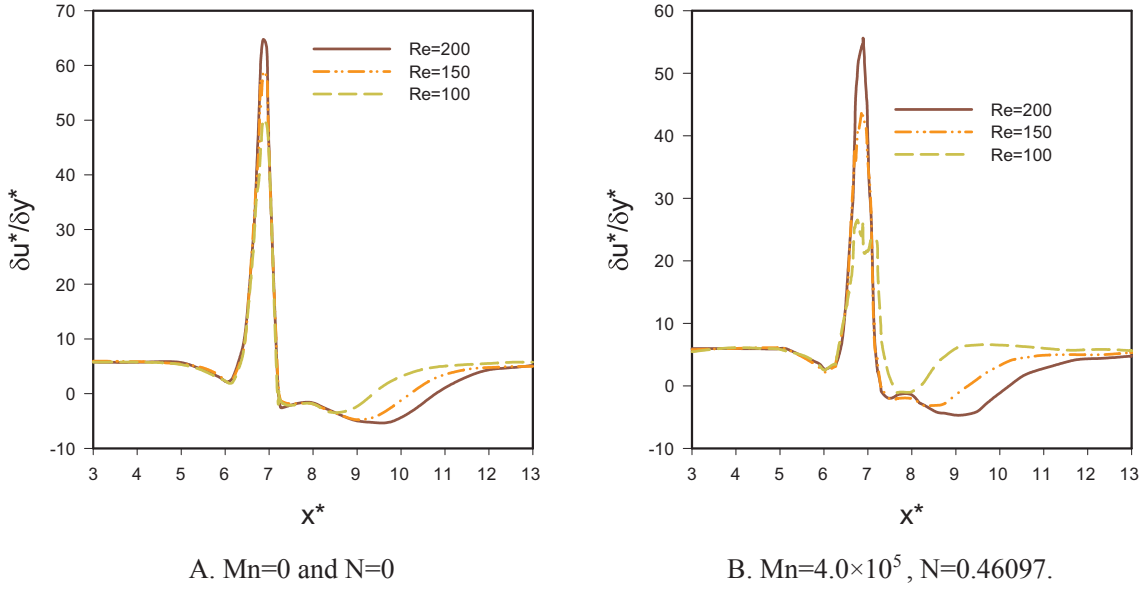


Fig. 11. Dimensionless velocity gradient on the bottom wall for δ/h = 0.5 in (A) Mn = N = 0 and (B) Mn = 4.0 × 10^5, N = 0.46097.

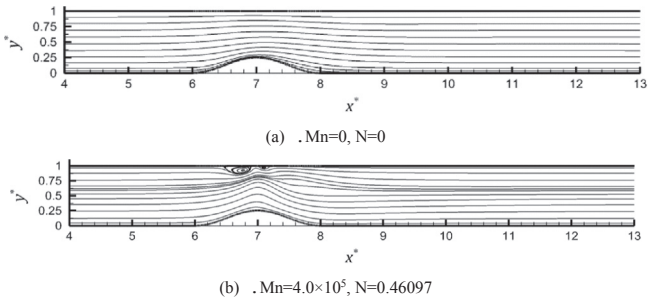


Fig. 12. Streamlines for δ/h = 0.25 and Re = 100.

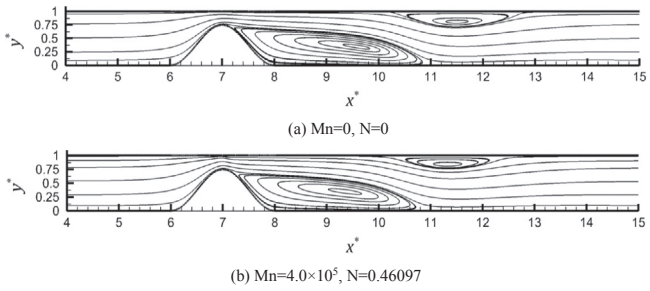


Fig. 14. Streamlines for δ/h = 0.75 and Re = 100.

mark of velocity gradient is changed two times.

In contours of (Fig. 14), the changes of streamlines are shown for duct with 75% (δ/H = 0.75) constriction without and with magnetic field. By increasing the constriction of duct, fluid jet becomes possible. It means that, by crossing the fluid from the section with protuberance and in the area after that, due to the increment of cross section, fluid is affected by strong velocity and pressure gradients and according to (Fig. 14), the vortexes are created with higher velocity components. In state with no magnetic field, two vortexes are created downstream. The vortex created behind the protuberance is because of the existence of

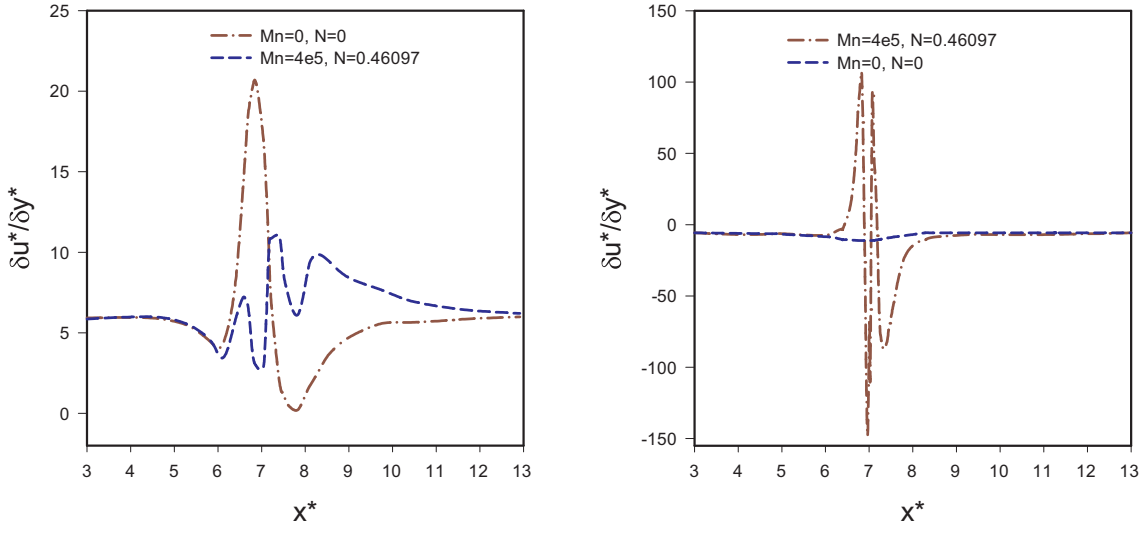


Fig. 13. Dimensionless velocity gradient on (A) bottom wall and (B) top wall in δ/h = 0.25 and Re = 100.

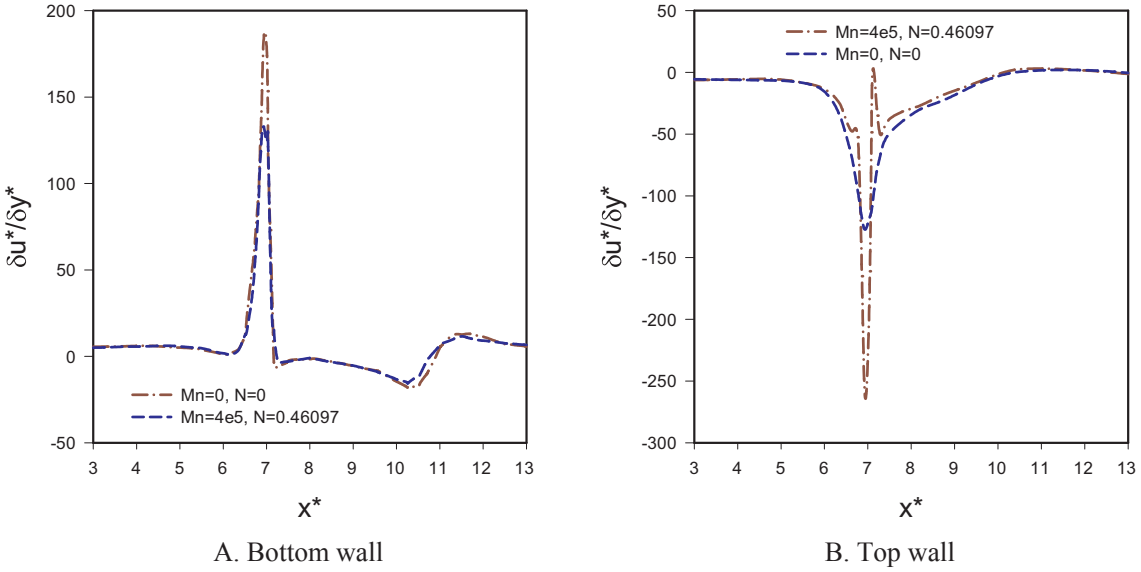


Fig. 15. Dimensionless velocity gradient on the bottom and top walls for δ/h = 0.75 and Re = 100.

strong positive pressure gradients. The other vortex has been created in $x^* = 10.5$ to $x^* = 12.5$ near the top wall and it is because of flow deviation toward the bottom wall. By applying the magnetic field, fluid accelerates toward the bottom wall of duct and the magnetic force decreases the pressure gradients behind the protuberance and reduces the vortex behind the constriction. Due to the downward flow caused by applying the magnetic field, the vortex close to the top wall is reduced in $x^* = 10.5$ to $x^* = 12.5$.

In (Fig. 15), the distribution of dimensionless velocity gradient on the bottom and top walls is shown in Reynolds number of 100 in δ/H = 0.75. Similar to studied figures for δ/H = 0.5 and δ/H = 0.25, the maximum velocity gradient and shear stress on the bottom wall takes place in the maximum constricted area and by applying the magnetic field, its value decreases. The notable point is that, the increase of δ/H causes more changes of velocity gradients and shear stress. The level of dimensionless velocity graphs in δ/H = 0.75 in comparison to δ/H = 0.5 and δ/H = 0.25, is increased for both of the walls. In general, the constriction with higher percentage has higher shear stress and applying magnetic force in δ/H = 0.75 has less effect on the reduction of flow separation of bottom wall. Furthermore, the constriction of duct with higher δ/H has considerable influence on the augmentation of shear stress on the top wall, and by increasing magnetic force, this behavior becomes more significant.

6. Conclusions

In the present paper, the effect of magnetic field caused by wire carrying electric current on the biomagnetic fluid flow has been investigated inside a duct with constriction. The distance of bottom wall of duct from the top wall has been lowered in the constricted area. The wire carrying electric current has been located near the top wall outside the duct. The results indicate that, by increasing the distance from the wire carrying electric current, which is the magnetic source, the effect of the magnetic field on the biomagnetic fluid flow decreases significantly. Applying the magnetic field with enough strength causes the recirculation areas downstream of constriction to become smaller. This issue limits the blood clotting (Thrombus). Furthermore, the magnetic field causes the reduction of shear stress on the wall of constricted zone. For instance, in 50% constriction and in $Mn = 4.0 \times 10^5$ and N = 0.4697, the maximum of shear stress decreases almost by 44% in comparison to the stress in state with no magnetic field. In addition, the presence of magnetic field significantly increases the wall shear stress near to the magnetic source.

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